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June 29, 1976
SUPERSEDING
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FEDERAL SPECIFICATION

WASH BOTTLE, LABORATORY

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types of wash bottles for laboratory use.

1.2 Classification.

1.2.1 Types, capacities and styles. Laboratory wash bottles covered by this specification shall be of the following types, capacities and styles as specified (see 6.2):

Type I - Borosilicate glass

Capacities:
500 milliliters (ml.)
1000 ml.

Type II - Polyethylene

Capacities:
250 ml.
500 ml.
1000 ml.

Styles:

A - Outlet tube through screwcap
B - Removable size outlet tube.
C - Integral side outlet tube.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

FSC 6640

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Federal Specifications:

L-P-390	-	Plastic Molding Material, Polyethylene Low and Medium Density.
DD-G-541	-	Glass (Laboratory).
DD-T-741	-	Tubing, Glass (Laboratory Use).
ZZ-S-751	-	Stoppers, Bottle, Rubber.
ZZ-T-331	-	Tubing, Rubber.
NNN-F-240	-	Flask, Laboratory.
PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-B-676	-	Box, Setup.

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

(Copies of Military Specification and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Materials. The materials used in the fabrication of the wash bottles shall be of high quality throughout, and suitable in every respect for the purpose intended.

3.1.1 Glass. All glass used in type I wash bottles shall be in accordance with type I, class A of DD-G-541.

3.1.2 Polyethylene. Polyethylene shall be in accordance with type I, grade 1 of L-P-390.

3.2 Construction. Type I flasks shall be as described in 3.2.1. Type II bottles shall be furnished in styles A, B or C, at the option of the manufacturer or supplier.

3.2.1 Type I.

3.2.1.1 Flask. The flass flask for type I wash bottles shall conform to NNN-F-240, type I, class 1, and shall be of the styles and sizes shown in table I for the capacity indicated. The walls of the flask shall be of sufficient thickness to afford the necessary strength and durability required for general laboratory use.

TABLE I. Flasks for type I wash bottles

Capacity of wash bottle	Applicable flasks of NNN-F-240, type I, class 1
500 ml.	Style I, size 2 or style II, size 1
1000 ml.	Style I, size 3 or style II, size 2

3.2.1.2 Insulation. The exterior of the neck and upper portion of the flask shall be covered with asbestos, cork or wicker to provide a suitable gripping surface and insulation against hot and cold liquids.

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3.2.1.3 Rubber bulb. Rubber bulbs shall be of a variety commercially available for use with wash bottles. Each bulb shall have a check valve to maintain pressure on the wash bottle liquid and prevent loss of pressure through the rubber bulb.

3.2.1.4 Rubber stopper. Rubber stoppers shall be in accordance with type B, grade I of ZZ-S-751 for the applicable sizes shown in table II. Each rubber stopper shall have two perforations in accordance with ZZ-S-751.

TABLE II. Rubber stoppers for type I wash bottles

Capacity of wash bottle	Rubber stopper number	Dimensions of stopper in millimeters (mm.)			
		Diameter of top	Diameter of bottom	Length	Diameter of perforations
500 ml.	6	32 ± 1	26 ± 1	25 ± 1	5.0 ± 0.5
1000 ml.	7	37 ± 1	30 ± 1	25 ± 1	5.0 ± 0.5

3.2.1.5 Rubber tubing. Rubber tubing shall be in accordance with ZZ-T-831, light wall, grade A or grade B. Two pieces of rubber tubing shall be required as connectors. One piece of the tubing, measuring 4 ± 1/4 inches, shall connect the rubber bulb (see 3.2.1.3) to the inlet tube (see 3.2.1.6.1). The other piece of tubing, measuring 2-1/8 ± 1/8 inches, shall connect the delivery tip (see 3.2.1.6.3) to the outlet tube (see 3.2.1.6.2).

3.2.1.6 Glass tubing. Glass tubing shall conform to type I of DD-T-741.

3.2.1.6.1 Inlet tube. The inlet tube shall be a short length of glass tubing, properly bent outward and down from the vertical axis to form an angle of 120 ± 6 degrees. The outside diameter of the tube shall properly fit the stopper perforation, and the tube shall terminate just below the stopper. The inlet tube shall be provided with a convenient means of releasing the pressure on the wash bottle contents, such as a hole in the tube, located for ease of finger operation. Both ends of the tube shall be fire polished.

3.2.1.6.2 Outlet tube. The outlet tube shall extend to within 5 mm. of the bottom of the flask. The tube shall be properly bent, and shall have an outside diameter that properly fits the stopper perforation. Both ends of the outlet tube shall be fire polished.

3.2.1.6.3 Delivery tip. The delivery tip shall be a short length of glass tubing. One end of the tip shall be fire polished, and the other end drawn out 2 to 3 centimeters (cm.) to a smaller diameter.

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3.2.2 Type II, style A (see figure 1). The bottle and liner shall be fabricated of polyethylene. The screwcap and outlet tube shall be fabricated of polyethylene or polypropylene.

3.2.2.1 Bottle. The bottle shall be wide- or narrow-mouthed, cylindrical, and of a diameter suitable for convenient laboratory handling. The mouth of the bottle shall be threaded to receive a screwcap. The wall of the bottle shall be of sufficient thickness to provide the required strength and durability for general laboratory use.

3.2.2.2 Screwcap and outlet tube. The outlet tube, with the screwcap in place on the bottle, shall extend from within 5 mm. of the bottom of the bottle upward through the center of the liner and the cap for a convenient distance, and then bend outward and downward from the vertical axis to form an angle which is less than the 90° angle shown in figure 1. The outlet tube shall terminate in a drawn-out tip of 2-3 centimeters (cm.), suitable for producing a small diameter stream. The wall of the outlet tube shall be of sufficient thickness to provide the necessary strength and durability required for general laboratory use. The inside diameter of the outlet tube shall be such as will allow a steady, controllable stream of liquid to be dispensed. The outlet tube shall be flexible throughout its entire length. The screwcap shall be properly threaded to mate and form a tight fit with the mouth of the bottle. Screwcap may be furnished with a liner, but liner is not required.

3.2.3 Type II, style B (see figure 1). The wash bottle, boss and outlet tube shall be fabricated of polyethylene. The screwcap shall be fabricated of polyethylene or polypropylene.

3.2.3.1 Bottle. The bottle shall be narrow-mouthed, cylindrical, and of a diameter suitable for convenient laboratory handling. The mouth of the bottle shall be threaded to receive the screwcap. The shoulder of the bottle shall be of sufficient thickness to provide the required strength and durability during general laboratory use.

3.2.3.2 Boss. The boss shall be integral with the bottle or shall form a tight friction fit with the hole in the shoulder of the bottle. A uniform hole running the entire length of the boss shall be provided to allow insertion of the outlet tube through the boss.

3.2.3.3 Outlet tube. The outlet tube shall be flexible throughout its entire length. The tube shall extend from within 5 mm. of the bottom of the bottle upward through and above the boss for a convenient distance, and then bend outward

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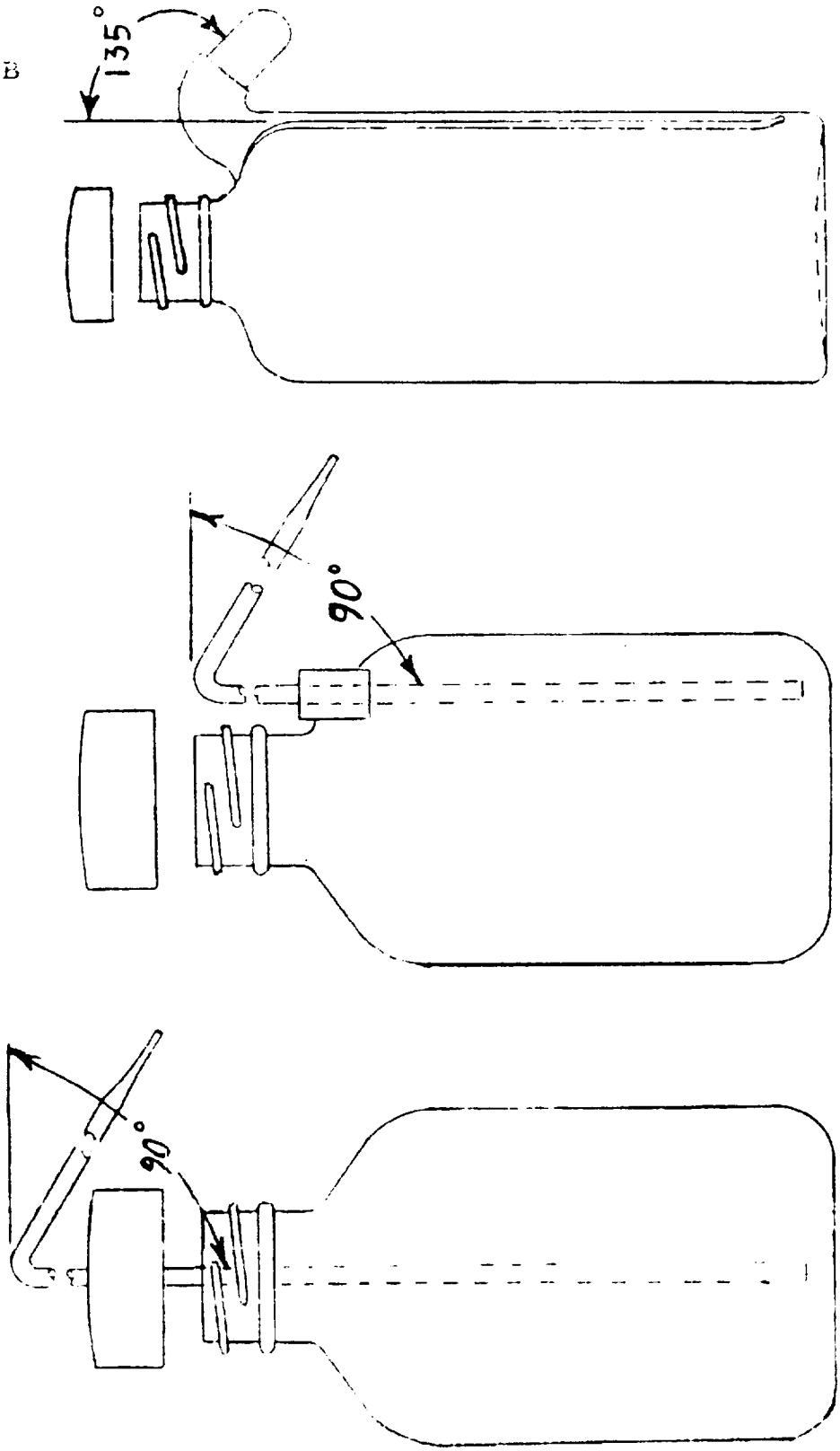


FIGURE 1. Wash bottles, type II

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and downward from the vertical axis to form an angle which is less than 90° angle shown in figure 1. The tube shall terminate in a drawn-out tip of 2-3 cm., suitable for producing a small diameter stream. The wall of the outlet tube shall be of sufficient thickness to provide the necessary strength and durability required for general laboratory use. The inside diameter of the outlet tube shall be such as will allow a steady, controllable stream of liquid to be dispensed.

3.2.3.4 Screwcap. The screwcap shall be properly threaded to mate and form a tight fit with the mouth of the bottle.

3.2.4 Type II, style C (see figure 1). The wash bottle and the outlet tube shall be fabricated of polyethylene. The screwcap shall be fabricated of polyethylene or polypropylene.

3.2.4.1 Bottle. The bottle shall be wide- or narrow-mouthed, cylindrical, and of a diameter suitable for convenient laboratory handling. The wall of the bottle shall be of sufficient thickness to provide the required strength and durability during general laboratory use. The mouth of the bottle shall be threaded to receive the screwcap.

3.2.4.2 Outlet tube. The outlet tube shall be molded in and shall be integral with the bottle. The tube shall open into the side of the bottle just above the junction of the side and the bottom. A portion of the tube outside the bottle shall be arced, and of curvature such that the top of the arc is below the top of the mouth of the bottle, and the end of the arc forms an angle of 135° ± 5° with the vertical axis. The tube shall terminate in a straight, rounded-off tip, or a tapered cone tip. A small orifice at the end of the tip shall allow for dispensing of a small diameter stream of liquid.

3.2.4.3 Screwcap. The screwcap shall be properly threaded to mate and form a tight fit with the mouth of the bottle.

3.2.5 Figure 1. The illustrations in figure 1 are intended to be typical.

3.2 Leakage (type II). The fit of the component parts of type II wash bottle shall be such that the wash bottle does not leak when tested as specified in 4.4.2.

3.4 Area for marking. Type I wash bottles shall have an area conveniently located on the flask, and suitably treated by blasting, etching, or other means, to provide for laboratory marking with pencil. The area shall show no change and shall remain satisfactory for marking after being subjected to the tests specified in 4.4.1.

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3.5 Workmanship. Workmanship shall be first class throughout. The wash bottles shall be free from defects listed in table V or other defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificate of quality. Certificates of quality, supplied by the manufacturer of the glass, polyethylene and polypropylene, may be furnished in lieu of actual performance of such testing by the contractor, provided lot ideni ty has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table III.

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TABLE III. Sampling for examination

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	2.5
Minor defects	II	4.0
For dimensional examination	S-2	4.0

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105. The acceptance number shall be zero. Test criteria shall be as shown in table IV.

TABLE IV. Sampling for testing

Characteristic	Requirement	Test procedure	Inspection level
Permanency of area for marking (type I)	3.4	4.3.1	S-1
Leakage (type II)	3.3	4.4.1	S-1

4.3 Examination. The wash bottles shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with, but shall not be limited to, the defects listed in table V:

TABLE V. Classification of defects

Category	Defect
Major:	
101	Wash bottle not free of cracks.
102	Wash bottle not free of scratches.
103	Wash bottle not free of blisters.
104	Wash bottle not free of deformation.
105	Walls of wash bottle not of uniform thickness.
106	Any component missing.
107	Bottle not free of contamination, such as oil, grease or foreign matter.
108	Cap does not fit bottle (type II).
109	Rubber stopper of incorrect size, or without two perforations (type I).
110	Rubber bulb does not contain a check valve (type I).
111	Ends of glass tubing not smooth (type I).

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TABLE V. Classification of defects (cont'd.)

Category	Defect
Minor	
201	Delivery arm angle incorrect (type II).
202	Identification marking not complete, correct or legible.

4.3.2 Dimensional examination. The wash bottles shall be examined for defects in dimensions. Any dimension not within the tolerances specified shall be classified as a defect.

4.3.3 Examination of preparation for delivery. An examination shall be made to determine compliance with the packaging, packing and marking requirements of Section 5. Defects shall be scored as specified in table VI. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with AQL of 4.0 expressed in terms of percent defective.

TABLE VI. Examination of preparation of delivery

Examine	Defect
Contents	Not as specified.
Container	Not as specified.
Marking	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Packaging and packing materials	Component missing or damaged.
Workmanship	Bulging or distortion of containers.

4.3.3.1 Examination of closure, waterproofing and banding of container. When the shipping container is required to comply with PPP-B-636, the examination for defects in closure, waterproofing or banding shall be in accordance with the appendix of the box specification.

4.4 Tests. Tests shall be conducted to determine compliance with the requirements of this specification. Distilled water and reagent grade chemicals shall be used throughout the tests.

4.4.1 Permanence of area for marking (type I). In performing the tests specified herein, a different flask shall be used for each test.

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4.4.1.1 Alkali test. The sample flask shall be immersed in one normal sodium hydroxide (1N NaOH) at room temperature, covered, and allowed to stand for 1-1/2 hours. The sample shall be removed, rinsed thoroughly in distilled water, and dried by rubbing vigorously with a cloth.

4.4.1.2 Hydrochloric acid test. The sample shall be immersed completely in concentrated hydrochloric acid (specific gravity 1.18) at room temperature, and allowed to stand for 1-1/2 hours. The sample shall be removed, rinsed thoroughly in distilled water, and dried by rubbing vigorously with a cloth.

4.4.1.3 Sulfuric acid test. The sample shall be immersed completely in concentrated sulfuric acid (specific gravity 1.84) at room temperature, and allowed to stand for one hour. The sample shall be removed, rinsed thoroughly in distilled water, and dried by rubbing vigorously with a cloth.

4.4.2 Leakage test (type II). The wash bottle shall be filled with distilled water, and the cap adequately tightened. The bottle shall be squeezed using normal hand pressure. Examination for evidence of leakage shall be made with the bottle in various positions.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B or C as specified (see 6.2).

5.1.1 Packaging quantities. The number of units of issue indicated in table VII shall be packaged in each unit, intermediate and exterior container, as applicable, for the required level of protection specified in the procurement document.

TABLE VII Packaging quantities

Type	Capacity	Unit package	Intermediate package	Exterior container
I	500 ml.	See table VII in NNN-F-240		
I	1000 ml.	See table VII in NNN-F-240		
II	250 ml.	Not required	12 units	288 units
II	500 ml.	Not required	12 units	144 units
II	1000 ml.	Not required	12 units	72 units

5.1.1.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

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5.1.2 Level A.

5.1.2.1 Unit package.

5.1.2.1.1 Type I. Type I wash bottles shall be packaged as specified in NNN-F-240.

5.1.2.1.2 Type II. Unit package shall not be required for type II bottles.

5.1.2.2 Intermediate package. The number of units of one style of type II bottles for the capacity as specified in table VII shall be packaged in a box constructed in accordance with PPP-B-566, PPP-B-636, class domestic or PPP-B-676. Closure shall be adequate to prevent accidental opening.

5.1.3 Level C. Unit shall shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall level A, B or C as specified (see 6.1).

5.2.1 Level A.

5.2.1.1 Type I. Type I wash bottles shall be packed as specified in NNN-F-240.

5.2.1.2 Type II. The number of units of one style only for the capacity as specified in table VII shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of PPP-B-636.

5.2.2 Level B. Type I flasks and type II bottles shall be packed as specified in 5.2.1.1 and 5.2.1.2, except that the exterior container for type II bottles shall conform to PPP-B-636, class domestic and the closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or purchase order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

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5.3.2 Military requirements.

5.3.2.1 Type I. Type I flasks shall be marked as specified in NNN-F-240.

5.3.2.2 Type II.

5.3.2.2.1 Intermediate package. Each intermediate package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown.

5.3.2.2.2 Exterior container. Exterior containers shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Intended use. The laboratory wash bottles shall be suitable for use as a hand-squeezed device in military medical laboratories and other general laboratories.

6.2 Ordering data. Purchasers should select the perferred options permitted herein and include the following information in procurement documents:

a. Title, number and date of this specification.
b. Quantity per intermediate and exterior container (type I only).
c. Types, capacities and styles (see 1.2.1).
d. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 This specification does not cover all types, classes, grades or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types normally procured to meet Federal requirements.

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